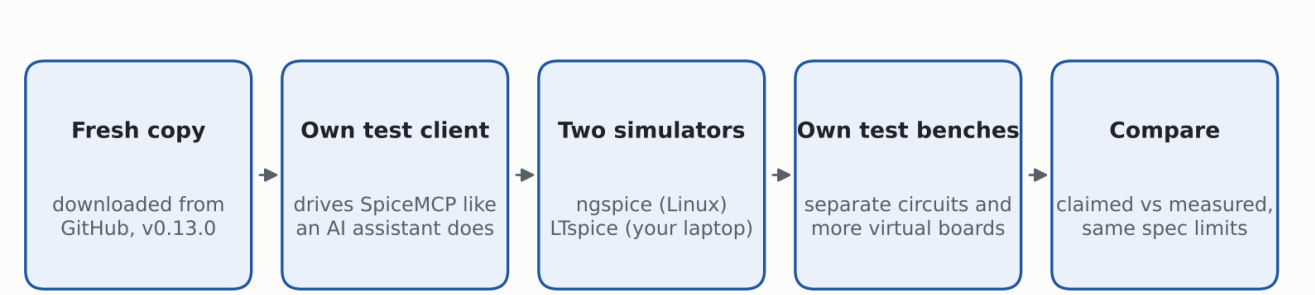
SpiceMCP 0.13.0

Independent verification: does it do what the report says?

Every claim in the engineering report (v0.13.0, commit 16c0ad0) was re-measured from a fresh download, using a separate test client, separate test circuits and, where possible, two different simulators. Then one circuit that is not in SpiceMCP's library, a three-op-amp instrumentation amplifier, was designed from a written spec to an orderable parts list using only the tool.



Verdict in one page

claim in the report	measured here	verdict
About 340 automated checks pass	339 pass, 41 skipped on Linux (they need LTspice). Windows + LTspice: pending	✔ holds
13 commands, each description short (limit 200 characters)	13 commands, longest 163 characters, no untyped inputs	✔ holds
Sallen-Key: −43 dB at 1 MHz with the real op-amp (−80 dB ideal)	-42.8 dB (ideal stand-in: -80.0 dB)	✔ holds
LDO: 72° phase margin with the series resistor, ~2° without	72.2° with, 0.8° without	✔ holds
LDO: 39 mV dip on a 10× load step	38.5 mV	✔ holds
Op-amp: 87 dB, 10.6 MHz, 10 V/µs, 1.1 mA	87.5 dB, 10.0 MHz, 9.3 V/µs, 1.16 mA	≈ mostly
Buck: input step peaks at 5.7 V, back in spec after 124 µs, ~89% efficient	5.72 V, 124 µs, 89.5%	✔ holds
Buck: 100% of boards regulate (closed loop)	Output voltage: 100% of boards in spec (4.95 to 5.03 V). But after the tool's own tune-and-finish steps some boards exceed 50 mV ripple: 95% pass on its 40 boards, 92% on 100 fresh boards	≈ mostly
Yield figures for the other four circuits	Reproduced exactly by the tool; my larger re-check lands within the expected scatter, except the amplifier (66%, not 50%)	≈ mostly
Optimizer finds a working Sallen-Key in 10 of 10 runs, ~20 simulations	19 of 20 new starts within 60; 9 of 10 with the real op-amp from scratch	≈ mostly
Every drawing is the same circuit as the netlist, pin by pin	connections checked by eye; LTspice read-back pending	≈ mostly
Schematic readability: LDO 97, buck 100	LDO 76, buck 69 today: the scores were measured before the real op-amp and the controller were added, and never updated	✘ does not hold
The final report PDF in the repository	The copy committed at v0.13.0 cannot be opened (broken file structure). Rebuilding it from its script works.	✘ does not hold

Plain-language summary: the electrical results are real and repeat in a second simulator. The weak spots are bookkeeping (a broken PDF, stale scores), how sure the yield numbers are, and drawings of circuits the layout was never tuned for.

CHECK 1

Install, tests and the command list

A clean copy was downloaded and installed exactly as a new user would, and the commands were listed by connecting to SpiceMCP the way an AI assistant does.

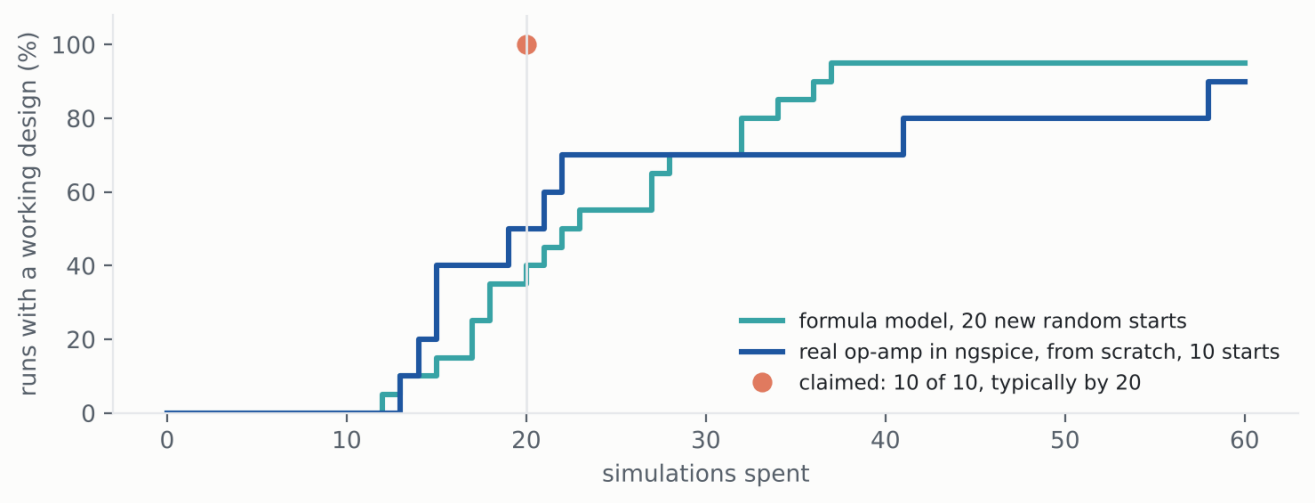
command	description length	inputs
check_setup	90 characters, 2 lines	0
simulate	146 characters, 2 lines	4
check_feasibility	156 characters, 2 lines	7
list_templates	155 characters, 2 lines	0
start_from_template	144 characters, 2 lines	4
start_optimization	131 characters, 2 lines	9
run_optimization	142 characters, 2 lines	2
evaluate_candidate	136 characters, 2 lines	3
query_run	142 characters, 2 lines	7
rollback_to_candidate	138 characters, 2 lines	2
finish_design	150 characters, 2 lines	8
generate_design_report	163 characters, 2 lines	8
render_schematic	138 characters, 2 lines	3

- **Tests:** 339 passed, 41 skipped on Linux; the skipped ones need LTspice. On your Windows laptop with LTspice: pending.
- **Commands:** 13, every description two lines and at most 163 characters. Every input has a declared shape, so the assistant never guesses a format.
- **Small inaccuracies:** the server's own header still says "11 tools", and *simulate* always asks for LTspice, so the one-off simulation command does not work on a machine with only ngspice (the design flow does).

CHECK 2

Does the optimizer find working part values?

The report says the Sallen-Key filter search now succeeds in 10 of 10 runs, typically after about 20 simulations. That was measured on ten particular random starts. I used twenty different ones, and also ran the harder real-op-amp version from scratch, with no hand calculation to start from.



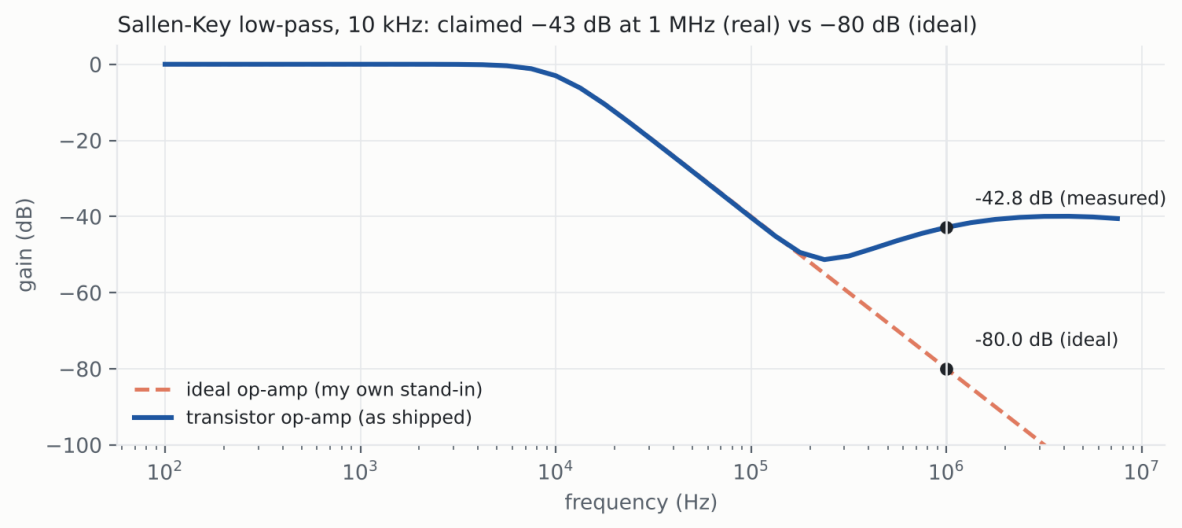
Share of runs that have found a design meeting every limit, against the number of simulations spent.

- Formula model, new starts: 19 of 20 succeed within 60 simulations; 8 of 20 within 20. The claimed median of about 20 holds; "10 of 10" is what those ten starts happened to give.
- Real op-amp in the simulator, no starting values: 9 of 10 succeed within 60 simulations, most within 15 to 25.
- RC filter: 20 of 20 within 20 simulations (claimed 10 of 10, about 6). Holds.
- **For you:** give a run 60 simulations, not 20, and check its status before trusting it. The tool tells you honestly when it has not found a design yet.

CHECK 3

Real op-amp, LDO stability and the closed-loop buck

These are the electrical claims that matter most on a bench. Each was re-measured with a test circuit written separately from SpiceMCP's own.



The real op-amp lets signal leak through above about 300 kHz; an ideal model hides it. Claim confirmed.

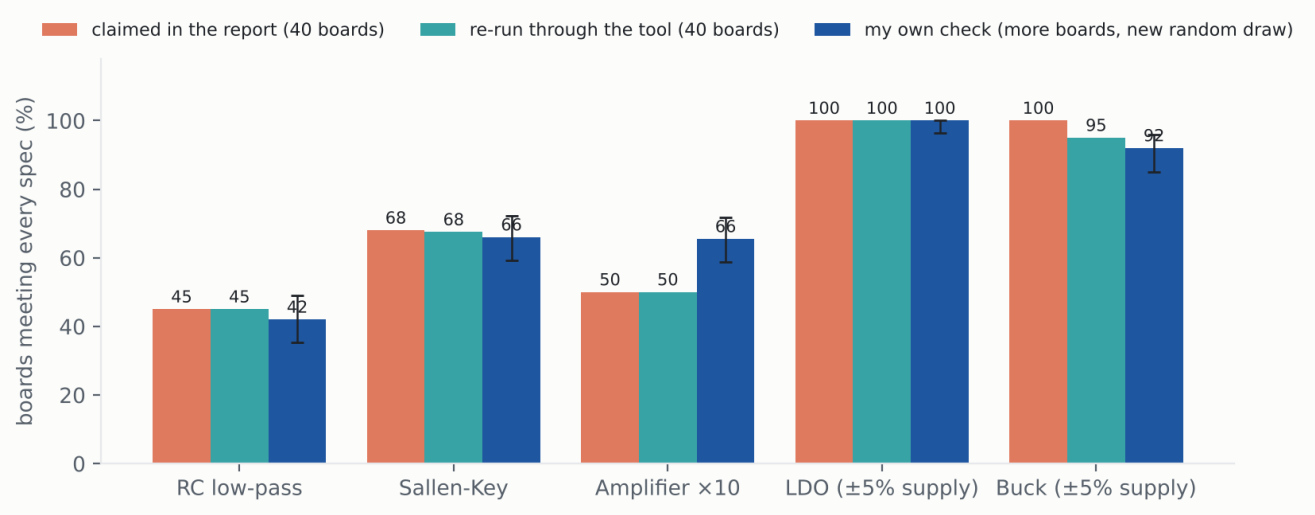
what	claimed	measured	verdict
Op-amp open-loop gain	87 dB	87.5 dB	✔ holds
Op-amp gain-bandwidth	10.6 MHz	10.00 MHz	≈ mostly
Op-amp slew rate	10 V/µs	9.3 V/µs	≈ mostly
Op-amp supply current	1.1 mA	1.16 mA	✔ holds
LDO phase margin, with 0.83 Ω series resistor	72°	72.2°	✔ holds
LDO phase margin, capacitor alone	2°	0.8°	✔ holds
LDO dip on a 10 mA → 100 mA step	39 mV	38.5 mV	✔ holds
Buck: peak when input jumps 11.4 → 12.6 V	5.7 V	5.72 V	✔ holds
Buck: back within ±2% after	124 µs	124 µs	✔ holds
Buck efficiency	about 89%	89.5%	✔ holds

- Gain-bandwidth and slew are 6–7% below the stated figures with my test load. The op-amp is sized for 10 MHz, so 10.6 MHz looks like a different load in the original measurement. Not important for any circuit here.
- **Gap:** the buck's control loop has no simulated stability check. Its 90° margin is a hand estimate, while the LDO and op-amp circuits get a real measurement. The input-step result above is good evidence the loop is stable, but it is not the same check.
- The buck overshoots more with the controller (5.72 V) than without (5.47 V), because the loop is slow. The report says so; worth knowing before you power anything sensitive from it.

CHECK 4

How many built boards would pass (yield)

Yield is estimated by simulating virtual boards with every part at a random value inside its tolerance. I re-ran the whole tool flow, then checked each finished parts list again with my own random draw and more boards.



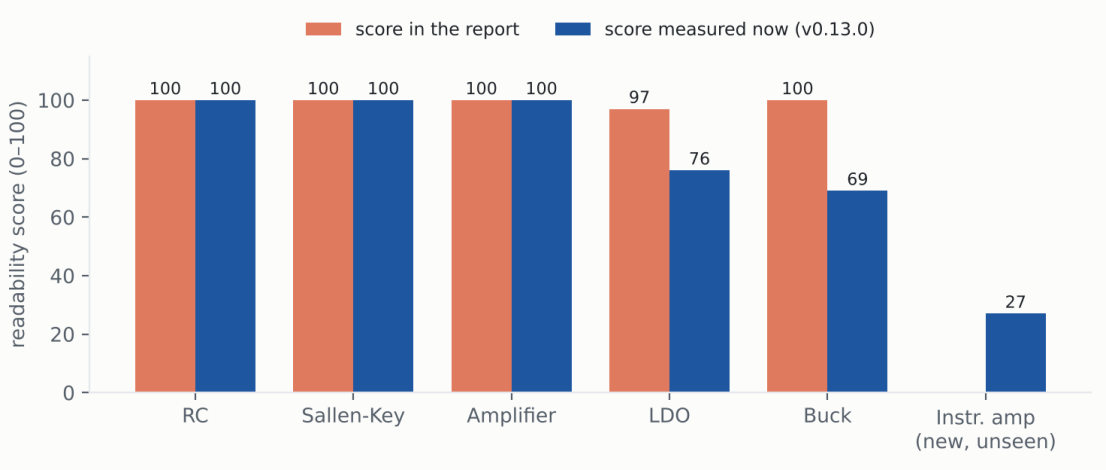
Black whiskers: the range the true yield lies in with 95% confidence, from my larger sample.

- **The tool's numbers reproduce exactly**: it always uses the same random draw, so running it again proves nothing new. That is why I made a fresh draw with more boards.
- RC, Sallen-Key and LDO: my check agrees with the claim.
- Amplifier: 66% on 200 boards, not 50%. With 40 boards a 15-point miss is just bad luck; the claim is too pessimistic, not wrong in direction.
- Buck: the report's 100% was measured at the hand-sized values. Through the tool's own tune-and-finish steps the inductor shrinks to 18 µH and 2 of 40 boards exceed 50 mV ripple (95%). My 100-board check: 92%; every failure is ripple of 50.1 to 53.8 mV, while the output voltage itself stays in spec on every board, so the regulation claim holds.
- **Wording to fix:** each answer says "40 runs resolve yield to about 2%". 40 boards give steps of 2.5%, but the uncertainty is ±8 to ±15 points. The engineering report's own "about ±8%" is the honest figure.
- **Subtle bug:** for the LDO the ±5% supply tolerance is applied to a source that the first test then sweeps anyway, so that part of the check does nothing. The LDO still passes when I apply the supply correctly, so the result stands.

CHECK 5

Automatic schematics

Two separate questions: is the drawing the same circuit as the netlist (a must), and is it easy to read (nice to have)?



Readability score: 100 minus points for crossings, wires through parts and similar.

- **Same circuit:** checked by eye on the new instrumentation amplifier and the buck; the LTspice read-back is in the LTspice section.
- **Readability:** the RC, Sallen-Key and amplifier drawings still score 100, and the eight small test circuits 99–100. The LDO (76) and buck (69) fell when real devices and the controller were added, and the report kept the old numbers.
- The new instrumentation amplifier draws correctly but scores 27: long wires and five crossings (picture on the next page). Three op-amps with cross-coupled feedback are beyond what the placer was tuned for, so "works on circuits it was never tuned for" holds only for small circuits.

NEW CIRCUIT

An instrumentation amplifier, designed with the tool

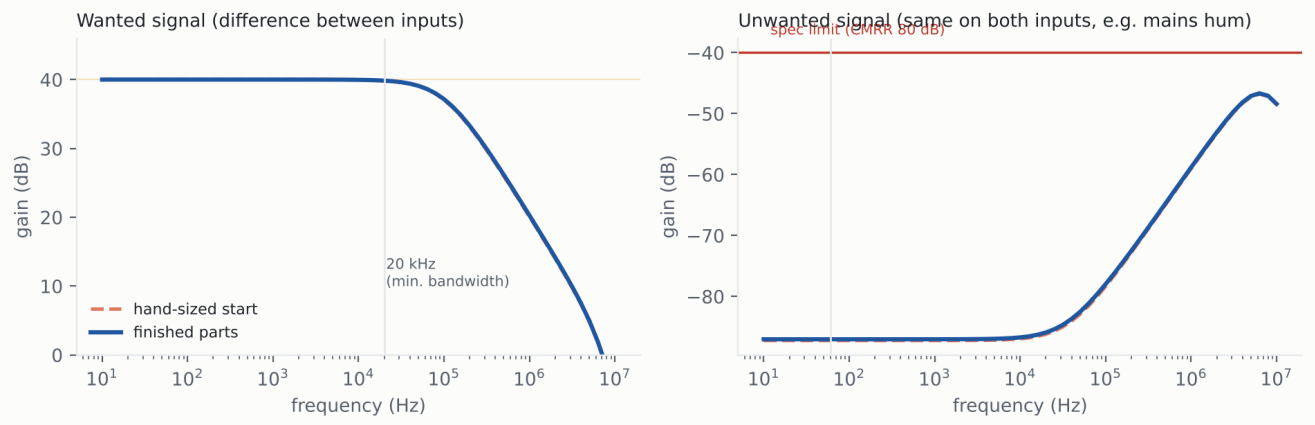
Not in SpiceMCP's library, so every step went through the general-purpose commands, the way an assistant would use them.

The spec, as an engineer would write it

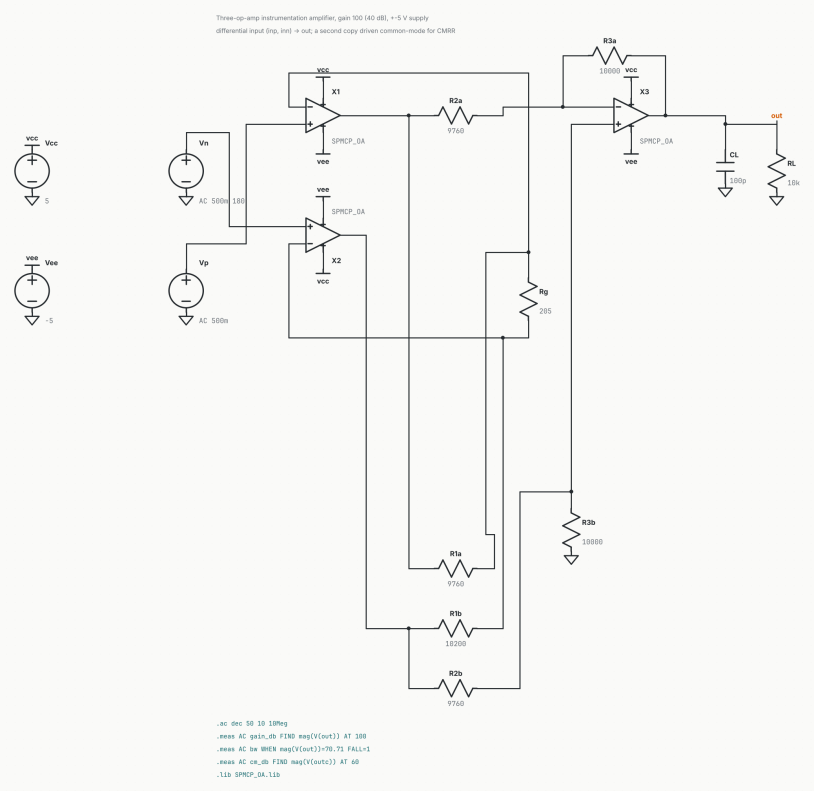
- Amplify the difference between two sensor wires by **100 (40 dB ± 0.1 dB)**.
- Bandwidth at least **20 kHz**.
- Reject a signal common to both wires (mains hum) by at least **80 dB at 60 Hz**.
- ±5 V supply, 10 kΩ and 100 pF load, stable (phase margin ≥ 45°). Standard 1% resistors.

What I did, in the tool's own commands

- **list_templates**: no instrumentation amplifier, so a custom design is needed.
- **start_from_template** + **query_run(netlist)** on the amplifier template, only to copy the transistor-level op-amp model. There is no command that hands it out directly.
- Wrote the circuit: three op-amps, seven resistors as tunable values, a second copy driven with the same signal on both inputs to measure hum rejection, and a loop-break test for stability. **check_feasibility** and **start_optimization** accepted it.
- **evaluate_candidate** with textbook values (all 10 kΩ, gain resistor 200 Ω), **run_optimization** (30 simulations), **finish_design** with 1% and with 0.1% resistors.



Sand band: the allowed gain window. Red line: the limit for the unwanted signal.



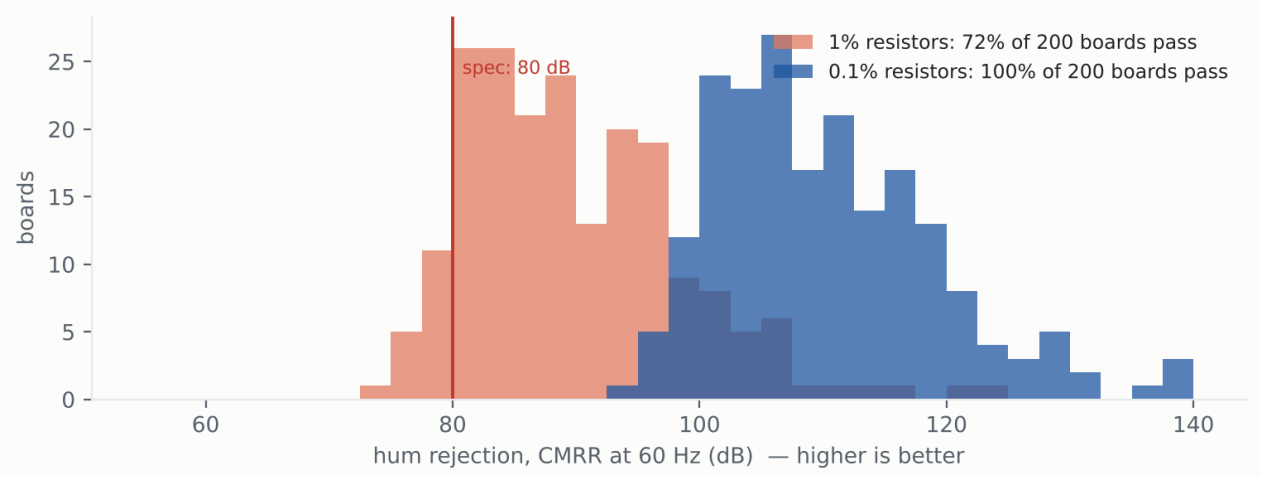
The schematic SpiceMCP drew for the finished amplifier. It is the right circuit, but untidy (score 27).

Parts list

part	value	tolerance	role
RG	205 Ω	0.1%	sets the gain
R1A	9.76k Ω	0.1%	feedback, input stage
R1B	10.2k Ω	0.1%	feedback, input stage
R2A	9.76k Ω	0.1%	difference stage in
R2B	9.76k Ω	0.1%	difference stage in
R3A	10k Ω	0.1%	difference stage feedback
R3B	10k Ω	0.1%	difference stage to ground
X1–X3	op-amp		any ±5 V op-amp, 10 MHz GBW class

Before and after

	hand-sized start	finished, standard values
gain	40.02 dB	40.00 dB
bandwidth	102 kHz	104 kHz
hum rejection (CMRR) at 60 Hz	127 dB	127 dB
phase margin	69°	70°
boards passing, 1% resistors	–	72% of 200
boards passing, 0.1% resistors	–	100% of 200



Each bar counts virtual boards. Hum rejection depends on how well four resistors match, so it is the spec that suffers from tolerance.

- **The tool got it right where it matters:** it confirmed the textbook design meets every spec, turned it into standard values and showed that 1% resistors are not good enough for 80 dB (only 72% of boards pass), while 0.1% resistors give 100%. That is the real-world answer: use 0.1% parts or a matched resistor network.
- **Where a human would do better:** the optimizer, told only to widen bandwidth, de-matched the resistor pairs for +3% bandwidth, dropping nominal hum rejection from 127 dB to 96 dB (still inside spec; snapping to standard values happened to restore it). An engineer would keep the pairs equal. Ask for both goals, or fix the pairs, when using it.
- finish_design with 1% resistors reported 48% yield on its 40 boards; my 200-board check of that same parts list gave 53%. Consistent.
- **Rough edges met on the way:** no command returns the op-amp model; the stability-test parts (1 kH, 1 kF) raise two false "impractical value" warnings; the design report on ngspice has no plots.

CHECK 6

The same checks in LTspice on Windows

SpiceMCP is meant for LTspice. Everything above was repeated on your laptop with LTspice, from a separate fresh copy.

- Pending: the LTspice run on your laptop had not reported back when this version was built. This page will be filled in when it does; everything above is from ngspice.

WHAT TO FIX

Discrepancies, most important first

None of these changes an electrical conclusion. Each is small to fix.

#	issue	suggested fix
1	docs/final/SpiceMCP_Engineering_Report.pdf at v0.13.0 is a broken file (two copies written over each other). The copy shared in the project is the same broken file.	Rebuild with make_pdf.py and add a test that opens the PDF.
2	Readability scores in the report (LDO 97, buck 100) predate steps 6 and 8.	Recompute from the current drawings, or label them by version.
3	Buck 100% yield holds at hand-sized values, not after tuning (95%).	Say which, or keep ripple margin when tuning shrinks L.
4	"40 runs resolve yield to about 2%" overstates precision; the random draw is fixed.	Report a confidence range; allow a seed.
5	LDO supply tolerance does nothing in the swept first test.	Move the read-out point with the supply, as done in this check.
6	Buck has no simulated loop-stability test.	Add a loop-break test like the LDO's.
7	Custom op-amp circuits need the op-amp model copied from a template run.	Return the model from list_templates or a resource.
8	simulate always uses LTspice; ngspice runs only through the design flow.	Honour the backend option there too.
9	Large circuits draw correctly but untidily (instrumentation amplifier: 27).	Group mirrored stages; a known limit is fine if stated.
10	Op-amp GBW/slew 6–7% below the stated 10.6 MHz / 10 V/µs.	State the test load.

How to repeat this

- Scripts: docs/verification/scripts/ (README inside). Raw results: docs/verification/results/.
- Rebuild this PDF: python docs/verification/make_report.py